

Application for Workshop/ Conference/ Lecture Support from IUCAA, Pune
(Application should be sent via email to meetings@iucaa.in)

Title of the Event: **Himalayan School for Training in Astronomy Research (HiSTAR)**

Date of the Event: **June 8-12, 2026 (summer school dates)**

Nature of the Event:

Sr. No.	Nature of Event	Please Tick
1	Workshop in a focused area	-
2	Meeting on a focused subject	-
3	Regional meetings, for e.g. NEMA, HMA, etc.	☒
4	International/National meetings	-

Venue: Applied Science Cluster, School of Advanced Engineering, UPES, Dehradun, Uttarakhand and Dolphin (PG) Institute of Biomedical & Natural Sciences (DIBNS), Dehradun

Local Coordinator, with email and phone number:

Dr. Nitesh Kumar

Email: nitesh.kumar@ddn.upes.ac.in

Phone: +91-8742930496

Dr. Asheesh Raturi

Email: aasheeshraturi@dolphininstitute.in

Phone: +91-7302546025

Prof. Hum Chand

CUHP, Dharamshala, HP

IUCAA Coordinator, with email and phone number:

Dr. Anupam Bhardwaj

Email: anupam.bhardwaj@iucaa.in

Phone: +91-9811650632

Other organizers:

1. Dr. Prince Sharma
2. Dr. Balendra P. Singh
3. Dr. Arka Chatterjee
4. Dr. Prashant Rawat

Rationale:

The Himalayan region of India offers some of the most pristine observing conditions in the world: high altitude, low atmospheric turbulence, dark skies, and low water vapor. With major national observatories such as ARIES (Nainital), the Indian Astronomical Observatory (Hanle), the region hosts frontline astronomical research facilities of India. Despite these strengths, formal training opportunities for students and early-career researchers in observational, computational, and instrumentation-based astronomy remain limited. This is particularly serious in hilly regions of **Uttarakhand, Himachal Pradesh, Jammu and Kashmir, and Ladakh** regions, where very few faculties in Universities are associated with astronomy research, education, and outreach. More importantly, undergraduate and post-graduate students do not have sufficient resources as well as opportunities to get involved in this exciting field of basic sciences.

To address this gap, we propose the **Himalayan School for Training in Astronomy Research (HiSTAR)**, a high-impact (potentially) annual, introductory educational program in research in astronomy and astrophysics, that will be designed to train participants with modern skills essential for pursuing research careers in astronomy and astrophysics. This regional school will primarily target students from hilly regions, but depending on the accessibility of where the school is organized, participation of University students from Punjab, Haryana, Delhi, and UP will also be possible, subject to availability of seats and resources.

The first proposed summer school will target many undergraduate and postgraduate students in local universities, introducing students to basics of astronomy and astrophysics, real-world astronomical datasets as well as basic data analysis and advanced machine learning applications. These schools aim to demystify research in astronomy and inspire the next generation of researchers from the region.

The HiSTAR is a training program that differs from the other regional HMA activity which targets only PhD/Postdoc/Early career researchers in A&A.

Objectives of the Event:

- Train early-career students (advanced UG, MSc, *first-year* PhD) in modern astronomy research techniques.
- Provide hands-on exposure to optical and infrared observational methods.
- Develop strong computational skills relevant to current astronomical research.
- Connect students with national facilities operating in the Himalayan region.
- Promote long-term scientific collaboration among Indian institutes.

- Strengthen human resource development for future national astronomy missions and facilities.

Target Audience: Final year BSc students, (in Physics / Applied Physics / Electronics / Astronomy / Applied Mathematics), M.Sc. students (in Physics, Applied Physics, Mathematics, Applied Mathematics, Electronics, or Astronomy) or B.E./ B.Tech. (final year) and / M.E./ M.Tech. students in physics/computer science, as well as first year University PhD. students.

Proposed Format: The first proposed school will be of (tentatively) 5 days. It is expected that longer duration schools will develop in future once the logistics and sufficient exposure is guaranteed.

Proposed Format (5 Days):

- **Morning Sessions:** Lectures by experts (Fundamentals of astronomy, The Sun, Stellar evolution, Compact Objects, Star clusters, Variable stars, Galaxies and ISM, Gravity and Cosmology, Multiwavelength Astronomy, Telescopes and Instrumentations)
- **Afternoon Sessions:** Hands-on training (handling basic astronomical data, python basics, data mining, color-magnitude diagrams, light curves, galaxy morphology etc).
- **Evenings:** Handling telescopes, Amateur astronomy training and observation sessions.
- **Day 5:** Summary, Q&A, and Valedictory.

Expected Outcomes:

The HiSTAR program is expected to generate a strong and sustainable impact on astronomy education and research capacity across the Himalayan and adjoining regions. By providing structured, hands-on training in observational techniques, data analysis, computational tools, and modern research methodologies, the school will equip undergraduate and postgraduate students with foundational skills necessary for pursuing advanced studies in astronomy and astrophysics. Participants will gain direct exposure to India's premier Himalayan observatories, fostering early familiarity with national research infrastructure and encouraging future telescope usage and proposal submissions. The program is also anticipated to broaden the scientific community in states such as Uttarakhand, Himachal Pradesh, Jammu & Kashmir, and Ladakh by creating a pipeline of trained students who may later contribute to local universities as researchers, educators, and outreach leaders. Additionally, HiSTAR will promote new collaborations between institutions, enhance regional scientific literacy, and inspire a new generation of students to engage with basic sciences, ultimately strengthening India's human resource base for upcoming large-scale astronomy missions and facilities.

Total number of participants expected (this should include resource persons and local persons as well students etc): 60

Mode of selection of participants: Participants will be selected through an online registration form that includes a short statement of interest. Priority will be given to students from physics and astronomy background from the Himalayan region (Uttarakhand, J&K, Himachal, Ladakh) as well as neighbouring northern states such as Punjab, Haryana, Delhi NCR, and UP.

Number of Outstation participants: 25

Estimated participation	Local	Other Parts of India
Speakers and Organisers	4	6
Students Participants	20	25
Faculty participants	5	0

Details of the speakers and faculty participants who require travel support:
(tentative list, speakers/resource persons may change, subject to availability)

Sr. No.	Name	Affiliation	Designation with Pay level as per 7th CPC	Travel request
1.	Prof. Hemwati Nandan	GU	Professor pay level 14	5000 (INR, road/train)
2.	Prof. Hum Chand	CUHP	Professor pay level 14	10,000 (INR, road/train)
3.	Prof. H. P. Singh	DU	Retired Professor IUCAA Associate	5000 (INR, road/train)
5	Dr. Ramesh Chandra	KU	Professor pay level 14	5000 (INR, road/train)
4.	Dr. Kaushal Sharma	FSL-UP	IUCAA associate	5000 (INR, road/train)

All resource persons are IUCAA associates. The travel allowances will be provided as per IUCAA norms.

Day-wise Tentative Schedule: This schedule may change depending on the availability of the invited speakers.

Himalayan School for Training in Astronomy Research (HiSTAR) – Summer School 2026

Tentative Programme Schedule

Dates - 8 - 12 June, 2026

**Venue - Applied Science Cluster, School of Engineering, UPES - Bidholi, Dehradun, UK.
And Dolphin PG Institute, Dehradun.**

The five-day workshop will follow a structured format:

- **Morning/afternoon Sessions (each day):**

09.00 am - 09.30 am : Registration [First day only]
09.30 am - 10.15 am : Inauguration [Day 1]/Science Talk
10.15 am - 11.00 am : Science Talk
11.00 am - 11.30 am : Tea Break
11.30 am - 12.15 pm : Science Talk
12.15 pm - 01.00 pm : Science Talk
01.00 pm - 02.00 pm : Lunch Break
02.00 pm - 02.45 pm : Talks/Projects
02.45 pm - 03.30 pm : Project & Softwares/hands-on-sessions
03.30 pm - 04.00 pm : Tea Break
04.00 pm - 05.30 pm : Individual/Group Projects

- **Evening Session (on specific days):**

Telescope observation session at UPES and Dolphin Institute (weather permitting) for direct exposure to sky-based observations and data acquisition.

- **Day 4 and 5:**

The lectures on Day 4 and Day 5 will commence in **Dolphin Institute**. Final lectures and hands-on session in the morning, followed by closing remarks and certificate distribution before lunch.

Coordinator(s) :

Dr. Nitesh Kumar - UPES

Dr. Asheesh Raturi - Dolphin PG Institute

Prof. Hum Chand - CUHP

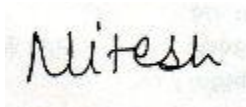
Prof. Anupam Bhardwaj
IUCAA

Projected Budget:

SN	Expense Item	No. of person	Cost Per Person (Rs)	Amount Per Day (Rs)	Total Number of Days	Total Amount (Rs)
1	Accommodation for resource** persons (UPES Guest house)	6	1100	6600	6	39,600
2	Accommodation for (outstation) student participants (DINBS Hostels – subsidized)	25	150	3750	6	22,500
2a	Bus travel from/to DINBS to/from UPES for student participants***			0	5	25000
3	Travel allowance for outstation regional participants^	25	2000	50000		50,000
4	Local transport for resource person (Airport/train station pick up and drop)	6	1000	6000	2	12,000
5	Food – Breakfasts (Outstation resource persons + students)	31	100	3100	6	18,600
6	Food – Lunches (60 participants)	60	250	15000	5	75,000
7	Food – Dinners (6 +25 persons)	31	250	7750	6	46,500
8	Tea/snacks (Total number of teas = 60*2 times a day)	120	50	6000	5	30,000
9	Stationary (pen, pad, badge etc. for 60 persons)	60	150	9000	1	9,000
10	Any others – Travel for resource persons (details in earlier page)^					30,000
11	Miscellaneous	100	(20 + 30)	5000	1	10,000
	Contingency (public lecture in UPES/Dolphin) [Tea + Snacks]					5,000
	Grand Total					3,73,200
	Support from Local Host or any other source (in what form): Guest house accommodation is provided by the host institute (UPES) for the invited resource persons. Seminar hall and local logistics are covered.					**39,600
	Student's transport from hostel to Venue (DINBS to UPES)					***25,000
	^Travel Allowances will be provided as per IUCAA norms.					
	Request from IUCAA					3,08,600

Any other details: The host institutes will ensure the necessary logistical support for the smooth conduct of the workshop, including access to computer labs, internet connectivity, and audio-visual facilities. Efforts will also be made to facilitate outreach by encouraging participation from nearby institutions.

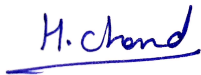
Signature of Authorized Personnel



(Dr. Nitesh Kumar - UPES)



(Dr. Asheesh Raturi - Dolphin PG Institute)



(Prof. Hum Chand - CUHP)



(Prof. Anupam Bhardwaj, IUCAA)